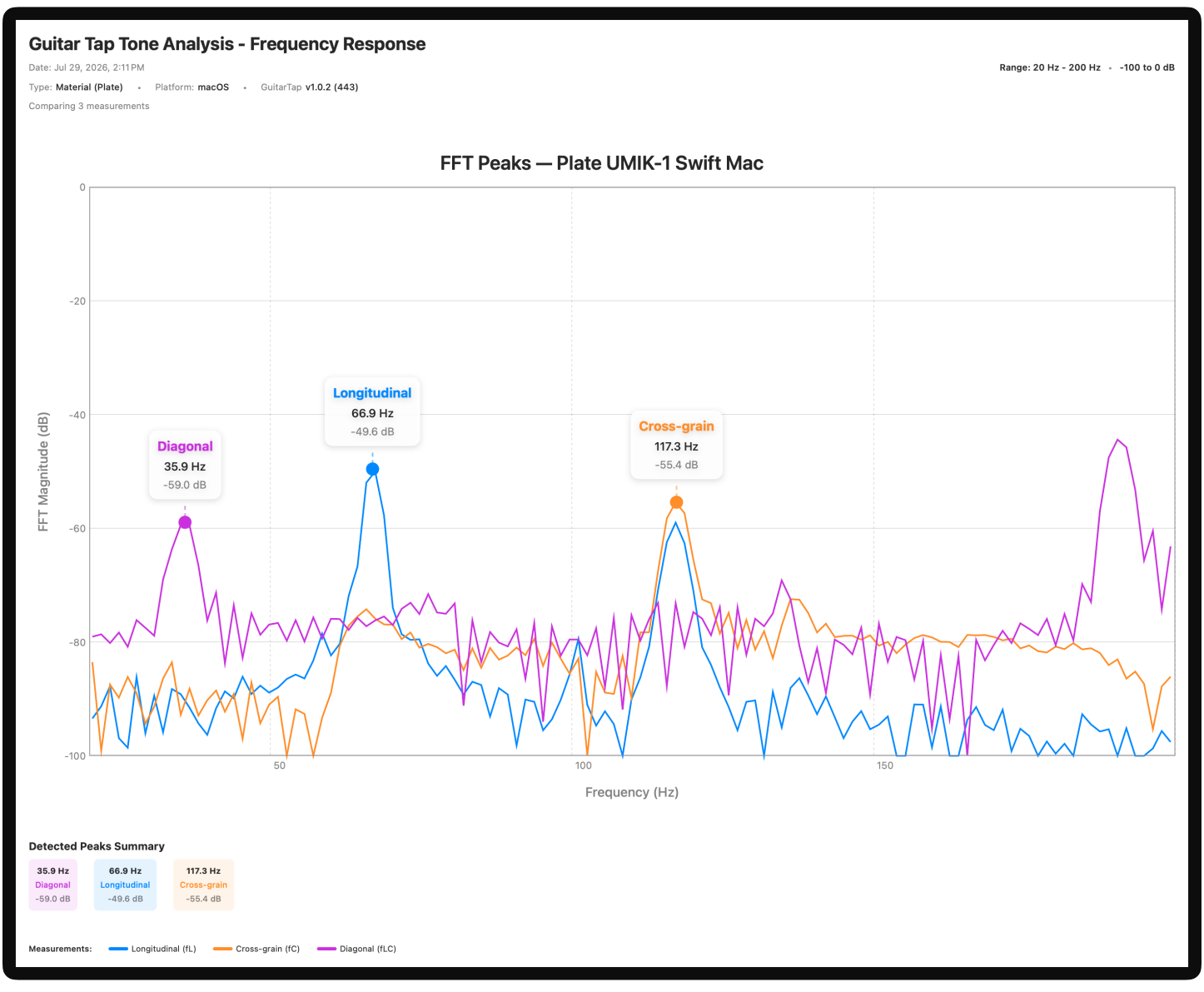
GuitarTap

Tap Tone Analysis Report

Jul 29, 2026, 2:11 PM

Measurement Name: Plate UMIK-1 Swift Mac
Type: Material (Plate)
Frequency Range: 20.0 Hz – 200.0 Hz
Microphone: Umik-1 Gain: 18dB · calibrated (7108913)

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
35.9 Hz	-59.0 dB	D1	8.0	Diagonal (fLC)
66.9 Hz	-49.6 dB	C2	15.3	Longitudinal (fL)
117.3 Hz	-55.4 dB	A#2	20.0	Cross-grain (fC)

Sample Dimensions

Length: **557.50 mm** Width: **220.50 mm** Thickness: **4.85 mm**
Mass: **208.0 g** Density: **0.349 g/cm³**

Body Dimensions

Body Length (a): **490 mm** Lower Bout Width (b): **368 mm**
Panel Stiffness: **f_vs = 60 (Classical Top)**

Gore Target Thickness

2.81 mm

Plate Properties

Speed of Sound (L): **4174 m/s**
Speed of Sound (C): **1144 m/s**
Young's Modulus (L): **6.08 GPa**
Young's Modulus (C): **0.46 GPa**
GLC (Shear Modulus): **0.350 GPa**
Cross/Long Ratio: **0.075**
typical: 0.04–0.08

Specific Modulus (L): **17.4** (Fair)
Specific Modulus (C): **1.3** (Very Good)
Radiation Ratio (L): **12.0**
Radiation Ratio (C): **3.3**
Long/Cross Ratio: **13.3**
typical: 12–25

Overall Quality: **Good**

Three-Tap Measurement Process:

- **1. Longitudinal (fL) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (fC) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. Diagonal (fLC) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.